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Submission date	September 21, 2026
Research topic	Utilizing Photovoltaics and Wireless Power Transfer to Advance Space-Based Solar Power

Title

Utilizing Photovoltaics and Wireless Power Transfer to Advance Space-Based Solar Power

Introduction

With increasing demands for electricity worldwide, sustainable and efficient sources of energy have been more important than ever before. [1] Solar photovoltaic systems are a renewable form of energy source, yet terrestrial solar panels are affected by day-night effects, absorption of the atmosphere, and presence of water. [2] SBSP involves installing massive photovoltaic systems in space that would generate much higher continuity in comparison to solar power installed on the surface of the earth. [3] The produced electricity from space can be transmitted to earth stations using microwave technology and then converted back into electricity. [3] Yet, such a process would require innovations in high-efficiency photovoltaics, energy conversion, wireless power transmission, beam control, and energy reception on the surface. [4]

In order to achieve such an objective, this study aims to explore how developments in photovoltaic technology and wireless power transmission can increase efficiency of space-based solar power for electricity generation on earth.

Research questions

1. What are the ways through which improvement in photovoltaic technology can help increase the efficiency of energy production by space-based solar power systems?

2. What are the determinants that affect the efficiency of wireless energy transmission from space-based solar power systems to Earth?

3. How could photovoltaic technology and wireless energy transmission work together to improve the efficiency of space-based solar power systems?

Literature review

The technology of space-based solar power (SBSP) has been proposed as a possible means to harness solar power in outer space and transmit the same down to earth in order to generate electricity. The basic principle of the system includes capturing the solar energy through photovoltaics, transforming the electricity produced from the same into a form that can be transmitted wirelessly, transmitting the energy down to earth through beaming, and then eventually transforming this energy into electricity. [4] This strategy seeks to overcome some of the shortcomings of solar power on the ground because solar energy collected in outer space does not depend on the day-night transition as well as climate conditions. [5]

Therefore, photovoltaic systems are integral to SBSP as the efficiency of photovoltaic cells determines the amount of electrical power that can be obtained from the incoming solar rays. Photovoltaic technologies such as multijunction III-V solar cells, perovskites, and silicon have been named by the U.S. The Department of Energy has some key research areas for space photovoltaics. [6] Previous studies conducted by NASA have also pointed out the significance of factors such as photovoltaic efficiency, radiation resistance, specific power, and temperature. [8] These results indicate that gains in efficiency of photovoltaic energy conversion processes may help reduce the quantity of solar collection area needed, among other resource needs, to produce a certain amount of energy. [7]

Another equally important element of SBSP would be wireless power transmission since the power created from space would need to be transmitted to the Earth using wireless technology. Both radio frequency and light-based technologies have been considered by NASA in its studies for transmission purposes. Microwave transmission plays a vital role because it is possible to use large antennas for forming the beam shape and beam direction in microwave communication systems. [8] Technological issues such as improved beam pointing accuracy, power to microwave conversion efficiency, microwave energy to electrical energy conversion efficiency, and reduction in the size and weight of the electronic components have been established by JAXA. [8]

The efficiency of wireless energy transmission is based not only on the process of transmission but also on several stages of the conversion of the energy. The evaluation of the Solar Power Satellite (SBSP) conducted by NASA reveals significant energy losses at the stage of solar energy collecting and conversion into the microwave energy. This means that improvement of each stage might be insufficient. [9] Previous studies by NASA too have underscored the progress made in power generation, power management, power beaming, and power conversion from beamed power as critical areas of development for SBSP technology.

The combination of PV power production along with wireless power transmission is crucial for the performance of the SBSP system as a whole. As noted by JAXA, the research on the SSPS involves efficient power production, transmission, and reception. [5] The Department of Energy of the United States describes a system where the energy from the sun captured by photovoltaic cells is converted into microwave or laser energy. [10] Nonetheless, as per NASA, SBSP is not

currently economically viable or feasible either, signifying a need for further research and development. [11] In summary, the improvement of SBSP requires not only energy production but also energy transmission through wireless technology. [8] Further research on efficient photovoltaic systems, energy conversion, beam control, and terrestrial reception may lead to increased efficiency of space-based solar power (SBSP). [8]

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Methodology

This study will use quantitative research for investigation of the impact that photovoltaic efficiency and wireless power transfer have on the overall efficiency of solar power satellites (SPS). The research will be limited to quantifiable indicators like efficiency of photovoltaic conversion, efficiency of energy transfer, energy loss, and the amount of generated electricity. Data will be collected mostly from scientific literature and reports.

Data collection

Data will be collected from peer-reviewed scientific journals, research papers, technical reports and other official sources, including NASA, JAXA and the United States Department of Energy. Data gathered will mainly concern photovoltaic effectiveness, wireless power transfer, microwave conversion, beam directing and energy receiving. Relevant numeric data on efficiency and energy loss will be gathered and organized for comparison.

Data analysis

The gathered numeric data will be compared in order to find out how the effectiveness of SBSP depends on different photovoltaic technologies and wireless power transmission methods. Percentages of efficiency and energy losses will be analyzed to find out what technological factors influence most of all the total energy provided to Earth. Tables and simple calculations can be used to show the result.

Relevance and reliability of sources

Sources will be considered according to their reliability, relevance, freshness and scientific authority. Priority will be given to peer-reviewed scientific sources and official organizations, including NASA, JAXA and other governmental research agencies. The information will be also checked in different sources to avoid the use of inaccurate data.

Ethical considerations

In this research, academic integrity will be preserved through appropriate referencing of all sources used and through non-plagiarism. There will be no distortion of information obtained from published data. There will be a clear distinction between known information, predicted values and new technologies.

Experimental design

Since the evaluation of SBSP technologies at large scale is very difficult, the experiment will mainly be based on secondary data. It will not involve building of the entire space system. This

is because a comparative modeling approach will be used to evaluate different energy conversion processes. These include photovoltaic energy conversion, energy transmission through the space channel and the energy reception process.

Possible outcome

It is expected that at the end of this experiment there will be a quantitative analysis of the effect that advancements in photovoltaic technology and wireless power transmission technology can have on the efficiency of SBSP.

Conclusion

Space based solar power (SBSP) can be used to provide constant power through renewable sources in space using solar energy. Efficiency in photovoltaic cells and the use of wireless energy transfer will reduce loss of energy and increase efficiency in the system. However, there are challenges involved that require research to be done before implementation on a larger scale.

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